AI Training: Roadmap for Beginners

AI Training for Beginners

Self-study training: use AI/LLMs safely in everyday life

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https://github.com/mgunthner/AI-Training

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AI Training: Roadmap for Beginners

Welcome! This roadmap shows you how the training is structured, which learning paths exist and in which order you work through the topics.

For whom: People without prior knowledge who want to use AI/LLMs in everyday life (free plans are sufficient for this). **Principle:** First understand, then use safely, then steer, then automate.

Learning paths

Choose the path that suits you. You can extend or switch paths at any time.

Path	For whom	Topics	Effort
Quick start	Everyone – mandatory part	00, 01, 02, 03, 04, 05, 08, 12 + quick test	about 3 hrs.
Standard	Regular users	00–08, 10–13, 22, 23 + tests A–C, F, optional capstone (25)	about 8–10 hrs. (+ 2–4 hrs. capstone)
Advanced	Tech enthusiasts, power users	all 24 (incl. 09, API-oriented) + all tests incl. final test, capstone (25) recommended	about 2 days (+ 2–4 hrs. capstone)

In the standard path, questions 14–15 (topic 09) are omitted from Test B; the pass mark is then 20/26.

How the training runs

Per topic you need about **25–30 minutes**:

1. **Read the cheatsheet** (about 10 min.)

2. **Do the exercise** (5–10 min., included in the cheatsheet)
3. **Extra exercises** (optional, as needed)
4. **Question?** → Open an issue in the GitHub repo (see below)
5. **Test** after each block

The total effort for the standard path is about 8–10 hours – freely divisible into small units.

Material overview

```
(your language folder, e.g. en/)
├── 00_introduction_handout       ← start here
├── 01_cheatsheets/
│   └── 01_what_is_ai … 25_capstone_project
├── 02_extra_exercises/
│   └── 01_extra_exercises … 25_capstone_extra_exercises
├── 03_tests/
│   ├── entrance_test
│   ├── test_A_foundation … test_F_final
│   ├── final_test
│   ├── self_assessment  (rubric for the self-assessment of practice
↪  tasks)
│   └── *_solutions  (one solutions file with reasoning for each test)
├── 04_trainer/
│   ├── question_template
│   └── faq  (grows with the questions)
├── 05_progress/
│   └── learning_progress  (checklist to tick off)
└── 06_capstone/  (after the final test)
    ├── project_instructions  (project instructions, start here)
    ├── example_portfolio  (fictional sample "Person A")
    └── checklist  (self-assessment, 24 points)
```

Tests

The tests make your learning progress visible – nothing more. You take them for yourself, without registration, and grade them yourself using the solutions file.

Test	Scope	Content	Grading
Entrance test	10 questions	Prior knowledge, self-assessment	not graded, shows your starting point
Test A–F (per block)	8–20 questions	Topics of the block	Pass mark: 75 %
Quick test: quick start	15 questions	Topics 01–05, 08, 12	Pass mark: 75 %

Test	Scope	Content	Grading
Final test	about 30 questions	all blocks, 60 % knowledge, 40 % practice	Pass mark: 75 %

Which question types you can expect: - Multiple choice (one or several correct answers) - True/False with reasoning - Scenario (“What do you do in this situation?”) - Practice task (e.g. improve a prompt, anonymize a text, recognize a risk) - Safety check (spot the mistake)

You assess practice tasks uniformly with the rubric in `03_tests/self_assessment` – criterion by criterion, stricter rather than lenient when in doubt.

If a test is not passed: No problem. The solutions file says which topics you should review – after that you can retake the test.

Extra exercises

For every topic 01–23 there is a file with 3–5 tasks (topic 24 is the glossary and has none). In addition, the capstone exercises (`02_extra_exercises/25_capstone_extra_exercises`) prepare the final project:

- **Three levels:** ★ entry, ★★ everyday, ★★★ challenge
- Everyday scenarios (email, vacation, office, learning, household)
- For each task: starting situation, task, hint – the solution approaches are deliberately at the end of the file
- Safety rule: only use invented or anonymized data
- Per block one optional transfer task: “Apply what you have learned to your own everyday life”

Questions? Support via GitHub issues

If you get stuck, open an **issue** in the GitHub repo – ideally with the template from `04_trainer/question_template`:

- Topic/cheatsheet number
- What did you want to achieve?
- What did you try? (insert prompt/result **anonymized**)
- Where are you stuck?
- Urgency

Important: No sensitive or personal data in issues – issues are public.

Frequently asked questions are collected anonymously in the FAQ (`04_trainer/faq`) and fed back into the cheatsheets.

Learning progress

In `05_progress/learning_progress` you tick off per topic: “read / practiced / test passed”. In addition, you can rate yourself (scale 1–5) – before and after, e.g. “I can write a good prompt”, “I recognize prompt injection”.

Topic overview and order

Work through the topics in numerical order. The “Path” column shows which learning path the topic is part of (S = quick start, St = standard, F = advanced; all topics also belong to all “smaller” paths before them, as far as listed).

Block	No.	Topic	Learning goal	Path
Start	00	Introduction handout	Know the safety rules, the process and how to start	S, St, F
A – Foundation	01	What is AI?	Roughly understand what an LLM is and what it is not	S, St, F
	02	Tokens & context window	Understand why chats “forget” and why limits exist	S, St, F
	03	Model selection & free plans	Choose the right model for the task	S, St, F
	04	Data privacy in everyday life	Know what belongs in a chat and what does not	S, St, F
B – Good results	05	Prompt basics	Write clear requests that are usable the first time	S, St, F
	06	Prompt techniques	Increase quality deliberately with proven patterns	St, F
	07	Prior knowledge & context	Give the AI the background knowledge it needs	St, F
	08	Hallucinations & fact-checking	Recognize and avoid wrong answers	S, St, F
	09	Temperature & sampling	Understand why answers vary and how to steer that	F
	10	Reasoning effort	Know when “thinking” pays off and what it costs	St, F
	11	Tools of the AI (tool use)	Understand what the AI can do besides text	St, F
C – Risks	12	Prompt injection	Recognize manipulation attempts via content	S, St, F

Block	No.	Topic	Learning goal	Path
	13	Guardrails	Know protective mechanisms and set your own guardrails	St, F
D – Technology & costs	14	API explained simply	Understand what an API is, without needing to program	F
	15	Cost control	Understand and keep costs and limits under control	F
E – Agentic systems	16	Agents	Understand what distinguishes an agent from a chatbot	F
	17	MCP	Understand how AI is connected to programs and data in a standardized way	F
	18	Skills	Know reusable capabilities and working instructions	F
	19	Subagents	Understand how tasks are distributed across several specialized AI helpers	F
	20	Advisor pattern	Know the interplay of a cheap “worker” and a strong “advisor”	F
	21	Running agents safely	Deploy agents in a controlled way and with a safety net	F
F – Final	22	Legal & compliance	Know the basic legal rules for everyday life (no legal advice)	St, F
	23	Everyday workflows & prompt library	Turn what you have learned into repeatable routines	St, F
	24	Glossary & further learning	Look up basic terms and keep learning on your own	F
Capstone	25	My personal AI workflow assistant	Turn what you have learned into a portfolio-worthy project (after the final test; grading: checklist in `06_capstone/checklist`)	St (optional), F (recommended)

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